

MY SQL MODULE END ASSIGNMENT

Name: Harikarthikeyan V

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E-Commerce Customer Churn Analysis Using MySQL

1. Project Title

E-Commerce Customer Churn Analysis Using MySQL

2. Project Overview

- This project focuses on analyzing customer churn in an e-commerce dataset using **MySQL**.
- The customer data was cleaned, transformed, and analyzed using SQL queries to identify customer behavior, churn patterns, payment preferences, order categories, complaints, satisfaction levels, and other factors related to customer churn.

The project also includes customer return data and combines it with customer churn information to identify customers who have both **churned and made complaints**.

3. Objective

The main objectives of the project are:

- To clean and prepare the e-commerce customer dataset.
- To handle missing values using appropriate imputation techniques.
- To identify and remove unwanted/outlier records.
- To standardize inconsistent categorical values.
- To transform existing columns into meaningful business categories.
- To analyze customer churn and active customer behavior.
- To identify patterns related to complaints, payment modes, order categories, and satisfaction.
- To analyze customer returns along with churn information.

- To use SQL queries for extracting meaningful business insights.
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4. Dataset Description

The project uses an **E-Commerce Customer Churn** dataset stored in the ecomm database.

The main table used in the project is:

Customer_churn

Important columns include:

- Customer ID
- Churn
- Tenure
- Preferred Login Device
- City Tier
- Warehouse To Home
- Preferred Payment Mode
- Gender
- Hour Spend On App
- Number Of Device Registered
- Preferred Order Cat
- Satisfaction Score
- Marital Status
- Number Of Address
- Complain
- Order Amount Hike From lastYear
- Coupon Used
- Order Count

- Day Since Last Order
- Cashback Amount

A second table named **customer_returns** was created to store customer return information.

5. Tools and Technologies Used

- **MySQL**
- **MySQL Workbench**
- **SQL**
- Relational database concepts
- Aggregate functions
- Conditional expressions
- Joins
- Subqueries
- Data cleaning and transformation techniques

6. Database and Table Setup

The database used for the project was:

USE ecomm;

The main customer table used was:

Customer_churn

The table structure and available records were verified using:

SHOW TABLES;

DESCRIBE customer_churn;

```
SELECT COUNT(*) AS Total Rows
```

```
FROM customer_churn;
```

Sample records were also reviewed using:

```
SELECT *
```

```
FROM customer_churn
```

```
LIMIT 10;
```

7. Data Cleaning

7.1 Identification of Missing Values

The dataset contained missing values in several numerical columns.

The following columns were checked:

- Warehouse To Home
- Hour Spend On App
- Order Amount Hike From lastYear
- Day Since Last Order
- Tenure
- Coupon Used
- Order Count

SQL aggregation was used to identify the number of NULL values in each column.

7.2 Mean Imputation

Mean imputation was performed for:

- Warehouse To Home
- Hour Spend On App
- Order Amount Hike From lastYear
- Day Since Last Order

The average value was calculated from the available records and used to replace NULL values.

Where required, the calculated mean was rounded to the nearest integer.

7.3 Mode Imputation

Mode imputation was performed for:

- Tenure
- Coupon Used
- Order Count

The most frequently occurring value was identified using GROUP BY, COUNT(), ORDER BY, and LIMIT.

7.4 Removal of Outlier Records

Records where:

Warehouse To Home > 100

were considered unwanted records according to the assignment requirement and were removed.

```
DELETE FROM customer_churn
```

```
WHERE Warehouse To Home > 100;
```

8. Data Standardization

Inconsistent categorical values were standardized to make the data more consistent for analysis.

Preferred Login Device

Phone was standardized to:

Mobile Phone

Preferred Order Category

Mobile was standardized to:

Mobile Phone

Preferred Payment Mode

COD was standardized to:

Cash on Delivery

CC was standardized to:

Credit Card

This standardization makes grouping and analysis more reliable.

9. Data Transformation

9.1 Column Renaming

The following columns were renamed:

Original Column	New Column
Prefered Order Cat	Preferred Order Cat
Hour Spend On App	Hours Spent On App

This improved the readability and consistency of the dataset.

9.2 Complaint Received Column

A new column named:

Complaint Received

was created from the original Complain column.

The transformation rules were:

- Complain = 1 → **Yes**
 - Otherwise → **No**
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9.3 Churn Status Column

A new column named:

Churn Status

was created from the original Churn column.

The transformation rules were:

- Churn = 1 → **Churned**
 - Otherwise → **Active**
-

9.4 Removal of Original Columns

After creating the transformed columns, the original:

- Churn
- Complain

columns were removed from the table.

This resulted in a cleaner and more understandable dataset.

10. Customer Churn Analysis

Several SQL queries were performed to analyze customer churn.

The analysis included:

10.1 Churned vs Active Customers

The number of customers in each churn status was calculated to understand the distribution between:

- Churned customers
 - Active customers
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10.2 Churned Customer Tenure and Cashback

The following metrics were analyzed for churned customers:

- Average tenure
- Total cashback amount

This helps understand the relationship between customer tenure, cashback benefits, and churn.

10.3 Complaints Among Churned Customers

The percentage of churned customers who submitted complaints was calculated.

This provides an indication of the relationship between customer complaints and churn.

10.4 Churn by City Tier and Order Category

Customers who had churned and preferred the **Laptop & Accessory** category were analyzed across City Tiers.

This helps identify the City Tier with the highest number of such churned customers.

10.5 Payment Mode Analysis

The preferred payment mode among active customers was identified.

This helps understand the payment preferences of currently active customers.

10.6 Order Amount Hike Analysis

The total Order Amount Hike From last Year was calculated for:

- Single customers
 - Preferred Order Category = Mobile Phone
-

10.7 Device Registration Analysis

The average number of registered devices was calculated for customers using **UPI** as their preferred payment mode.

10.8 City Tier Customer Distribution

The number of customers in each City Tier was calculated to identify the customer distribution across different tiers.

10.9 Coupon Usage by Gender

Coupon usage was analyzed by gender to identify the gender associated with the highest coupon usage.

10.10 Order Category Analysis

For each preferred order category, the following were calculated:

- Number of customers
- Maximum hours spent on the app

This provides a comparison of customer participation and app usage across product categories.

10.11 Credit Card and Satisfaction Analysis

Customers using **Credit Card** were analyzed based on their maximum Satisfaction Score, and the corresponding total order count was calculated.

10.12 Complaint and Satisfaction Analysis

The average Satisfaction Score of customers who complained was calculated.

This helps examine satisfaction levels among customers who reported complaints.

10.13 Customers Using More Than Five Coupons

Preferred order categories were identified for customers who used more than five coupons.

10.14 Top Three Categories by Cashback

The preferred order categories were ranked according to their average CashbackAmount, and the top three categories were identified.

10.15 Tenure and High Order Count Analysis

Customers with:

- Tenure = 10 months
- Order Count > 500

were analyzed to identify their preferred payment modes.

11. Distance Category Transformation

A new column named:

Distance Category

was created based on Warehouse To Home.

The classification was:

Warehouse To Home	Distance Category
≤ 5	Very Close Distance
> 5 and ≤ 10	Close Distance
> 10 and ≤ 15	Moderate Distance
> 15	Far Distance

A CASE statement was used to implement this transformation.

The resulting distance categories were then analyzed against customer churn status.

12. High-Order Customers Analysis

Customers satisfying all the following conditions were identified:

- Marital Status = Married
- City Tier = 1

- Order Count is greater than the average Order Count of all customers

A subquery was used to calculate the overall average order count.

This analysis helps identify high-order customers within a specific customer segment.

13. Customer Returns Analysis

A new table named:

Customer_returns

was created to store return information.

The table contains:

- Return ID
- Customer ID
- Return Date
- Refund Amount

Eight return records were inserted into the table as specified in the assignment.

14. Customer Returns and Churn Analysis

The customer_returns table was joined with the customer_churn table using:

Customer ID

An INNER JOIN was used to identify customers who:

- Made a return
- Churned
- Received a complaint

The final result combines return details with relevant customer information such as:

- Customer ID
- Return Date
- Refund Amount

- Tenure
 - Preferred Login Device
 - City Tier
 - Preferred Payment Mode
 - Gender
 - Preferred Order Cat
 - Satisfaction Score
 - Marital Status
 - Complaint Received
 - Churn Status
-

15. SQL Concepts Used

The following SQL concepts were used throughout the project:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- HAVING
- COUNT()
- SUM()
- AVG()
- MAX()
- ROUND()
- CASE
- UPDATE
- DELETE

- ALTER TABLE
 - ADD COLUMN
 - RENAME COLUMN
 - DROP COLUMN
 - CREATE TABLE
 - INSERT INTO
 - INNER JOIN
 - Subqueries
 - NULL handling
 - Data standardization
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16. Key Learning Outcomes

Through this project, the following skills were developed:

- Understanding relational database structure.
- Working with an existing MySQL database.
- Inspecting and validating table structures.
- Handling missing data using mean and mode imputation.
- Removing unwanted records.
- Standardizing inconsistent categorical values.
- Creating derived columns using SQL.
- Using conditional logic with CASE.
- Performing aggregation and grouping.
- Writing analytical SQL queries.
- Using subqueries for dynamic calculations.
- Joining multiple tables using common keys.
- Performing customer churn analysis using SQL.

17. Conclusion

The **E-Commerce Customer Churn Analysis** project demonstrates how MySQL can be used to transform raw customer data into structured and analysis-ready information.

The project covered the complete workflow from data cleaning and transformation to customer churn analysis and return-data integration. SQL queries were used to investigate customer behavior, complaints, payment preferences, order categories, satisfaction scores, cashback, distance categories, and churn status.

The final customer-return analysis further connected return information with customer churn and complaint status, providing a consolidated view of relevant customer records.

Overall, the project provided practical experience in **SQL-based data cleaning, transformation, analysis, aggregation, subqueries, and table joins.**